

Problem 25.8

Given

$$\bar{a}_{\overline{n}|} = n - 4 \quad \text{and} \quad \delta = 10\%,$$

find

$$\int_0^n \bar{a}_{\overline{t}|} dt.$$

Solution

Since $\delta = 10\%$, we have

$$\delta = 0.1.$$

For a continuous annuity,

$$\bar{a}_{\overline{n}|} = \frac{1 - e^{-\delta n}}{\delta}.$$

Using $\delta = 0.1$ and the given condition $\bar{a}_{\overline{n}|} = n - 4$,

$$\frac{1 - e^{-0.1n}}{0.1} = n - 4.$$

Thus,

$$1 - e^{-0.1n} = 0.1n - 0.4.$$

Rearranging,

$$e^{-0.1n} = 1.4 - 0.1n.$$

Now evaluate the integral:

$$\int_0^n \bar{a}_{\overline{t}|} dt = \int_0^n \frac{1 - e^{-0.1t}}{0.1} dt.$$

Since $1/0.1 = 10$,

$$\int_0^n \bar{a}_{\overline{t}|} dt = 10 \int_0^n (1 - e^{-0.1t}) dt.$$

Therefore,

$$\int_0^n \bar{a}_{\overline{t}|} dt = 10 \left[t + \frac{e^{-0.1t}}{-0.1} \right]_0^n.$$

Evaluate at the endpoints:

$$\int_0^n \bar{a}_{\overline{t}|} dt = 10 \left(n + \frac{e^{-0.1n}}{-0.1} - \frac{1}{-0.1} \right).$$

Since $1/0.1 = 10$,

$$\int_0^n \bar{a}_{\overline{t}|} dt = 10n + 100e^{-0.1n} - 100.$$

Substitute $e^{-0.1n} = 1.4 - 0.1n$:

$$\int_0^n \bar{a}_{\overline{t}|} dt = 10n + 100(1.4 - 0.1n) - 100.$$

Simplifying,

$$\int_0^n \bar{a}_{\bar{t}|} dt = 10n + 140 - 10n - 100 = 40.$$

Thus,

$$\boxed{40}.$$